

github.com/Bruno-LimaHG/IA-ATIVIDADE

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

IA-ATIVIDADE (New)

Watch Fork Star

Start coding with CodeSpaces
Add a README file and start coding in a secure, configurable, and dedicated development environment.
[Create a codespace](#)

Add collaborators to this repository
Search for people using their GitHub username or email address.
[Invite collaborators](#)

Quick setup — if you've done this kind of thing before
[Get up in Desktop](#) or [HTTPS](#) [SSH](#) <https://github.com/Bruno-LimaHG/IA-ATIVIDADE.git>
Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [gitignore](#).

...or create a new repository on the command line

```
echo "# IA-ATIVIDADE" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -m main
git remote add origin https://github.com/Bruno-LimaHG/IA-ATIVIDADE.git
git push -u origin main
```

...or push an existing repository from the command line

```
git remote add origin https://github.com/Bruno-LimaHG/IA-ATIVIDADE.git
git branch -m main
git push -u origin main
```

ProTip! Use the URL for this page when adding GitHub as a remote.

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IA-ATIVIDADE / deployment.yaml in main

Cancel changes Commit changes...

1 application: app/v1
2 kind: Deployment
3 metadata:
4 name: master-magic
5 labels:
6 app: master-magic
7 spec:
8 replicas: 1
9 selector:
10 matchLabels:
11 app: master-magic
12 template:
13 metadata:
14 labels:
15 app: master-magic
16 containers:
17 - name: master-magic
18 image: registry.k8s.io/http-ecp/v1
19 ports:
20 - containerPort: 80
21 resources:
22 requests:
23 cpu: 200m
24 limits:
25 cpu: 200m
26

Use `Ctrl` + `Shift` + `F` to toggle the `Tab` key moving focus. Alternatively, use `Esc` then `Tab` to move to the next interactive element on the page.

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Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

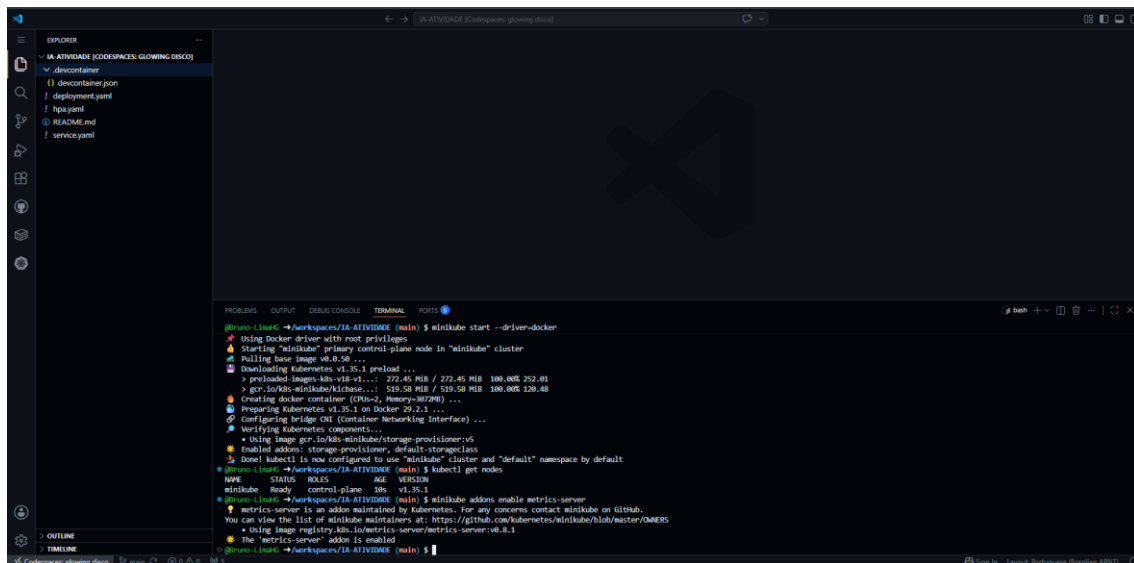
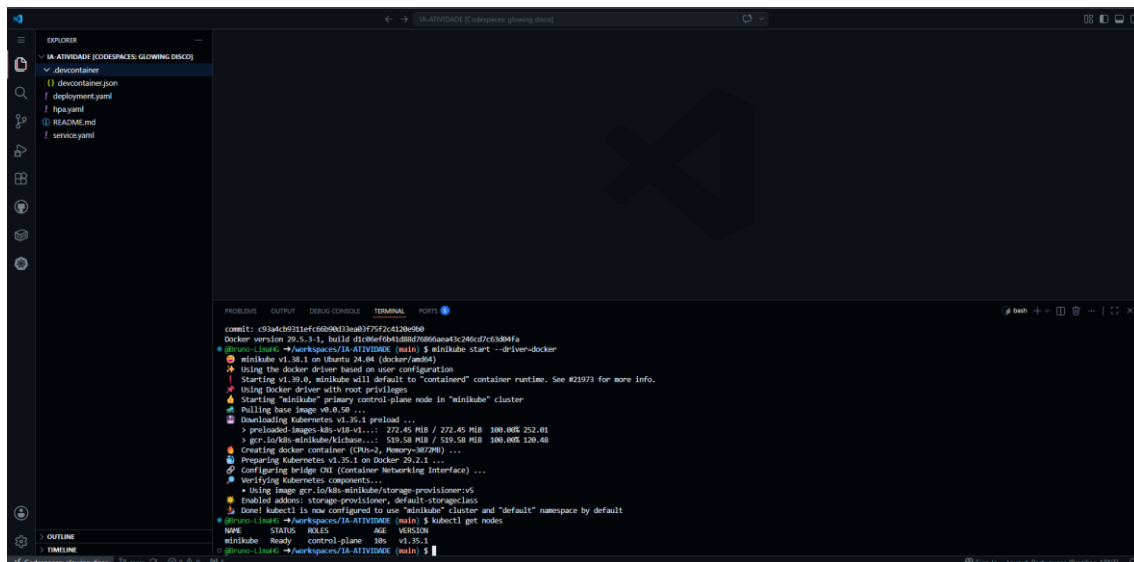
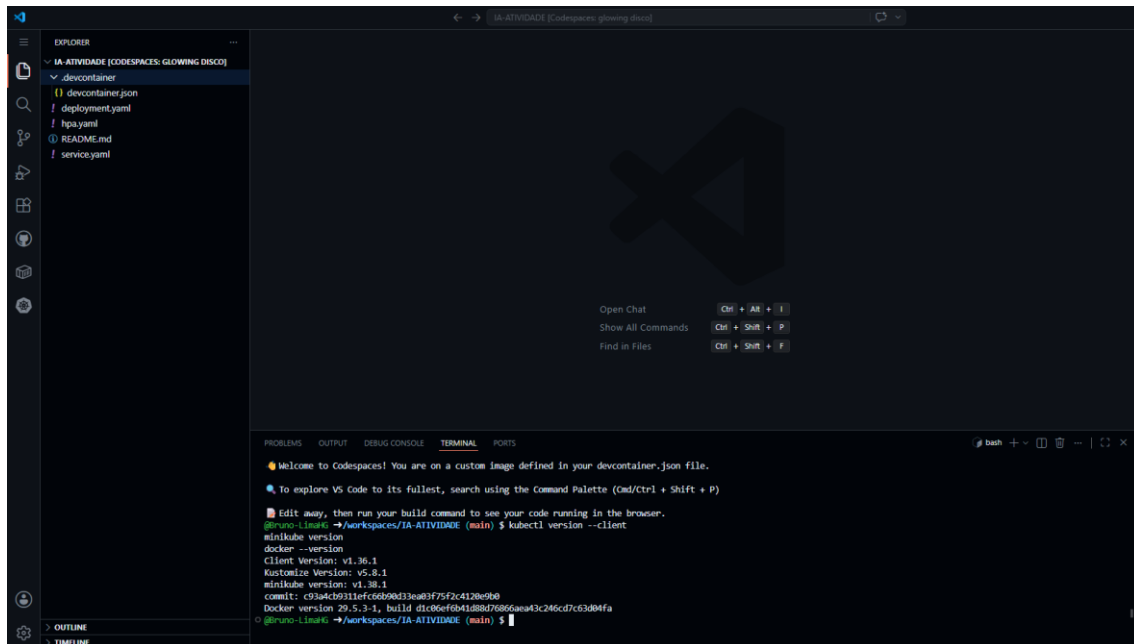
IA-ATIVIDADE / README.md in main

Cancel changes Commit changes...

1 # Atividade IA - Codespaces
2
3
4 Atividade pratica para demonstrar escalonamento horizontal com Kubernetes usando Github Codespaces, Kubectl, Deployment, Service e Infs.
5
6
7 ## Objetivo
8 Descrever como uma aplicação pode escalar o número de réplicas conforme a demanda aumenta.
9
10
11 ## Requisitos principais
12 - Codespaces.yaml: cria o aplicativo.
13 - deployment.yaml: cria o escalonamento.
14 - svc.yaml: configura o escalonamento automático.
15 - deployment/autoscaler.yaml: prepara o escalante do Codespaces.
16 |

Use `Ctrl` + `Shift` + `F` to toggle the `Tab` key moving focus. Alternatively, use `Esc` then `Tab` to move to the next interactive element on the page.

Attach files by dragging & dropping, selecting or pasting them.



```
EXPLORER
- IA-ATIVIDADE [CODESPACES G...]
  - decontainer
    - decontainer.json
    - deployment.yaml
    - hpa.yaml
    - README.md
    - service.yaml

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ minikube start --driver=docker
• Using image gcr.io/k8s-minikube/storage-provisioner:v5
• Enabled add-on: storage-provisioner, default-storageclass
• Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl get nodes
minikube Ready control-plane 10s v1.35.1
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ minikube addons enable metrics-server
! metrics-server is an add-on maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
• Using image registry.k8s.io/metrics-server/metrics-server:v0.6.1
• The "metrics-server" add-on is enabled
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl get pods -n kube-system
NAME READY STATUS RESTARTS AGE
coredns-7d78666f9-http 1/1 Running 0 67s
etcd-minikube 1/1 Running 0 72s
kube-apiserver-minikube 1/1 Running 0 72s
kube-controller-manager-minikube 1/1 Running 0 72s
kube-proxy-wd8s 1/1 Running 0 67s
kube-scheduler-minikube 1/1 Running 0 73s
metrics-server-bd48b608-h951 1/1 Running 0 45s
storage-provisioner 1/1 Running 1 (36s ago) 65s
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $
```

```
EXPLORER
- IA-ATIVIDADE [CODESPACES G...]
  - decontainer
    - decontainer.json
    - deployment.yaml
    - hpa.yaml
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• The "metrics-server" add-on is enabled
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coredns-7d78666f9-http 1/1 Running 0 67s
etcd-minikube 1/1 Running 0 72s
kube-apiserver-minikube 1/1 Running 0 72s
kube-controller-manager-minikube 1/1 Running 0 72s
kube-proxy-wd8s 1/1 Running 0 67s
kube-scheduler-minikube 1/1 Running 0 73s
metrics-server-bd48b608-h951 1/1 Running 0 45s
storage-provisioner 1/1 Running 1 (36s ago) 65s
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl wait --for=condition=available --timeout=10m deployment/metrics-server -n kube-system
deployment.apps/metrics-server condition met
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $
```

```
EXPLORER
- IA-ATIVIDADE [CODESPACES G...]
  - decontainer
    - decontainer.json
    - deployment.yaml
    - hpa.yaml
    - README.md
    - service.yaml

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
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• Using image registry.k8s.io/metrics-server/metrics-server:v0.6.1
• The "metrics-server" add-on is enabled
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl get pods -n kube-system
NAME READY STATUS RESTARTS AGE
coredns-7d78666f9-http 1/1 Running 0 67s
etcd-minikube 1/1 Running 0 72s
kube-apiserver-minikube 1/1 Running 0 72s
kube-controller-manager-minikube 1/1 Running 0 72s
kube-proxy-wd8s 1/1 Running 0 67s
kube-scheduler-minikube 1/1 Running 0 73s
metrics-server-bd48b608-h951 1/1 Running 0 45s
storage-provisioner 1/1 Running 1 (36s ago) 65s
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl wait --for=condition=available --timeout=10m deployment/metrics-server -n kube-system
deployment.apps/metrics-server condition met
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl top nodes
NAME CPU(cores) CPU% MEMORY(bytes) MEMORY%
minikube 128m 15% 128Mi 15%
@runo-linux: ~/workspace/IA-ATIVIDADE (main) $
```

```
EXPLORER
  IA-ATIVIDADE [CODESPACES: GLOWING DISCO]
    devcontainer
      devcontainer.json
      deployment.yaml
      hpa.yaml
      README.md
      service.yaml

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ minikube addons enable metrics-server
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl get pods -n kube-system
NAME READY STATUS RESTARTS AGE
coredns-7d78666f9-htbp 1/1 Running 0 67s
etcd-minikube 1/1 Running 0 72s
kube-apiserver-minikube 1/1 Running 0 72s
kube-controller-manager-minikube 1/1 Running 0 72s
kube-proxy-wd86 1/1 Running 0 67s
kube-scheduler-minikube 1/1 Running 0 71s
metrics-server-6d4b668-h991 1/1 Running 0 45s
storage-provisioner 1/1 Running 0 (36s ago) 68s
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl wait --for=condition=available --timeout=180s deployment/metrics-server -n kube-system
deployment.apps/metrics-server condition met
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl top nodes
NAME CPU(cores) CPU(%) MEMORY(bytes) MEMORY(%)
minikube 116m 12% 1200Mi 12%
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/avatar-magico created
service/avatar-magico-svc created
horizontalpodautoscaler.autoscaling/avatar-magico-hpa created
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $
```

```
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl get pods
kubectl get deployment
kubectl get service
kubectl get hpa

NAME READY STATUS RESTARTS AGE
avatar-magico-58dff698-p94bc 0/1 ContainerCreating 0 16s

NAME READY UP-TO-DATE AVAILABLE AGE
avatar-magico 0/1 1 0 16s

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
avatar-magico-svc ClusterIP 10.109.125.239 <none> 80/TCP 16s
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 3m11s

NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
avatar-magico-hpa Deployment/avatar-magico cpu: <unknown>/30% 1 3 1 17s
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $
```

```
EXPLORER
  IA-ATIVIDADE [CODESPACES: GLOWING DISCO]
    devcontainer
      devcontainer.json
      deployment.yaml
      hpa.yaml
      README.md
      service.yaml

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
horizontalpodautoscaler.autoscaling/avatar-magico-hpa created
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl get pods
kubectl get deployment
kubectl get service
kubectl get hpa

NAME READY STATUS RESTARTS AGE
avatar-magico-58dff698-p94bc 0/1 ContainerCreating 0 16s

NAME READY UP-TO-DATE AVAILABLE AGE
avatar-magico 0/1 1 0 16s

NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
avatar-magico-svc ClusterIP 10.109.125.239 <none> 80/TCP 16s
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 3m11s

NAME REFERENCE TARGETS MINPODS MAXPODS REPLICAS AGE
avatar-magico-hpa Deployment/avatar-magico cpu: <unknown>/30% 1 3 1 17s
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $ kubectl run -it --rm load-generator --image=busybox --restart=Never -- /bin/sh -c "while true; do wget -q -O- http://avatar-magico-svc; done"
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
@Bruno-LimaHG → /workspaces/IA-ATIVIDADE (main) $
```


EXPLORER

IA-ATIVIDADE [CODESPACES: GLOWING DISCO]

devcontainer

devcontainer.json

deployment.yaml

hpa.yaml

README.md

service.yaml

Preview README.md X

Aula Kubernetes IA - Codespaces

Atividade prática para demonstrar escalonamento horizontal com Kubernetes usando GitHub Codespaces, Minikube, Deployment, Service e HPA.

Objetivo

Observar como uma aplicação pode aumentar o número de réplicas conforme a demanda aumenta.

Arquivos principais

- deployment.yaml: cria a aplicação.
- service.yaml: cria o endereço interno da aplicação.
- hpa.yaml: configura o escalonamento automático.
- devcontainer/devcontainer.json: prepara o ambiente do Codespaces.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ minikube start --driver=docker
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.6.50 ...
minikube get nodes?
[!] Stopped minikube start --driver=docker
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  29s   v1.25.1
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/avatar-magico created
service/avatar-magico-svc created
horizontalpodautoscaler.autoscaling/avatar-magico-hpa created
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  30s   v1.25.1
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ minikube addons enable metrics-server
metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
Using image registry.k8s.io/metrics-server/metrics-server:v0.6.1
The "metrics-server" addon is enabled
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $
```

bash

bash

EXPLORER

IA-ATIVIDADE [CODESPACES: G...

devcontainer

devcontainer.json

deployment.yaml

hpa.yaml

README.md

service.yaml

Preview README.md X

Aula Kubernetes IA - Codespaces

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
kubectl apply -f hpa.yaml
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  30s   v1.25.1
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ minikube addons enable metrics-server
metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
Using image registry.k8s.io/metrics-server/metrics-server:v0.6.1
The "metrics-server" addon is enabled
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl wait --for condition=available --timeout=180s deployment/metrics-server -n kube-system
deployment.apps/metrics-server condition met
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl top nodes
NAME      CPU(%)    CPU(bytes)    MEMORY(%)
minikube  17%       8K            12%
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/avatar-magico unchanged
service/avatar-magico-svc unchanged
horizontalpodautoscaler.autoscaling/avatar-magico-hpa unchanged
@bruno-linux: ~/workspace/IA-ATIVIDADE (main) $
```

bash

bash